Deliverable: Requirements

Cohort 1, Team 8

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Methods for User Evaluation

Our team aimed to recruit 5-6 participants who could complete a task based user evaluation of our prototype. Task based user evaluation is based on 7 steps:

1. Recruit participants
2. Define a set of tasks
3. Prepare an environment
4. Plan how to run the tasks
5. Have participants complete the tasks
6. Compile the data
7. Make design recommendations

We recruited participants for user interviews on opportunity. This means that we simply asked people who we knew already or who were in our timetabled lab session. We aimed for 5-6 participants who could complete a task based user evaluation. The participants were all from the intended user group and also all students currently enrolled in the ENG1 module as required. Before engaging in any tasks the participants had to read through and complete the ethics form to show that they consent.

The task to be completed was for the user to play the game for 5 minutes and think aloud while doing so. All of the interviews were completed online on Discord. The users were sent a copy of the game and the relevant ethics and evaluation forms. Users were told to think aloud while playing the game so the interviewer could write down any issues they have or anything that worked as intended.

Data was collected from the interviews in two ways. The first was for users to think aloud while completing the task and for the interviewer to write down what they say. This data was stored in a google doc for each interview which is located on our team google drive. The second way was for users to complete a questionnaire on google forms after finishing the task which included questions about how difficult they found the game and any issues they might have had. This data was stored in the google form responses section which is located on our team google drive.

The full procedure for each interview is as follows:

1. Interviews are scheduled for a time which is suitable for the participant.
2. Users read and fill in the information sheets and consent forms provided to them by the interviewer.
3. Participants are sent a copy of the game for them to download
4. Participants must play the game and think aloud while doing so. As well as sharing their screen on discord if possible. The interviewer takes notes on what they say and do.
5. When the participants have finished playing the game they are sent a questionnaire made using google forms which they fill in.

Usability Problems in the Prototype System

Severity Rating Scale:
1. Cosmetic Problem (Green): This problem is making it slightly difficult to complete my task.
2. Minor Problem (Yellow): This problem is making it difficult to complete my task.
3. Major Problem (Orange): This problem is making it very difficult to complete my task.
4. Catastrophic Problem (Red): This problem makes my task impossible to complete.

Problem	**Severity Rating**
Clip through some walls	1
Game did not communicate objective well	2
Controls to get out text boxes was not intuitive	1
Interaction bug. Need to be moving to interact with objects	3
Not clear at all what to do when you get outside, no indication of what direction to go or what you are even looking for	4
Not clear what can be interacted with	2
No audio feedback when interacting with objects	1
Lots of sprites were reused	1
All events were in one place	1

To further improve the prototype system the interaction bug affecting object use should be resolved and clearer in-game cues should be introduced to guide player movement and decision making. This could include improved visual indicators, clearer prompts and more consistent feedback when actions are available. This helps players understand where to go and what to do at each stage of the game.

We made several changes to address issues discovered during and as a result of the user evaluation process. These included fixing interface bugs that led to events activating incorrectly, removing the abuse of repeated sprites to add visual variation and fixing collision problems where the player might clip through walls. In addition, event handling logic was redesigned by making sure all the events were in different locations instead of them being in one location. More adjustments were made to asset placement and interaction timing to ensure more smooth gameplay and consistent behaviour across levels.